Traps Lab - Ayaan Arif Nadeem - 30546

RISC-V assembly

output

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ call
12 13
HE110 World
```

BACKTRACE

1)Added def in defs.h

```
// printk.c
int             printk(char*, ...) __attribute__ ((format (printf, 1, 2)));
void            panic(char*) __attribute__((noreturn));
void            printkinit(void);
void            backtrace(void);
```

2)Added func to allow reading the s0 register in riscv.h

```
//way to read the s0 register
static inline uint64
r_fp()
{
  uint64 x;
  asm volatile("mv %0, s0" : "=r" (x) );
  return x;
}
```

3)implemented func in printk.c

```
void
backtrace(void)
{
  printk("backtrace:\n");
  uint64 fp = r_fp();
  uint64 top = PGROUNDUP(fp);

  while(fp < top) {
    uint64 ra = *(uint64*)(fp - 8);
    printk("%p\n",(void*)ra);
    fp = *(uint64*)(fp - 16);
  }
}
```

4)added func call to sys_pause in sysproc.c

```
uint64
sys_pause(void)
{
  int n;
  uint ticks0;

  backtrace();

  argint(0, &n);
  if (n < 0)
    n = 0;
  acquire(&tickslock);
  ticks0 = ticks;
  while (ticks - ticks0 < n) {
    if (killed(myproc())) {
      release(&tickslock);
      return -1;
    }
    sleep_prepare(&ticks);
    release(&tickslock);
    sleep();
    acquire(&tickslock);
  }
  release(&tickslock);
  return 0;
}
```

5)Added func call to panic in printk.c

```
void
panic(char *s)
{
  panicking = 1;
  printk("panic: ");
  printk("%s\n", s);
  panicked = 1; // freeze uart output from other CPUs
  backtrace();
  for (;;)
    ;
}
```

output

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ bttest
backtrace:
0x0000000080001ea6
0x0000000080001d82
0x0000000080001b02
```

```
root@DESKTOP-RBTBS7K:~/xv6-postmtlabs/xv6-labs-2026# addr2line -e kernel/kernel
0x0000000080001ea6
./kernel/sysproc.c:75
0x0000000080001d82
./kernel/syscall.c:146 (discriminator 1)
0x0000000080001b02
./kernel/trap.c:81
```